UTAC v2.0 – README FIRST

Overview & Orientation for Reviewers and Endorsers

Status: 2025-11-16 | Document Type: Corrected Version

Purpose of This Document

This document serves as the **first orientation** for reviewers and endorsers. It:

1. Separates **empirically validated results** (UTAC v2.0 Main) from **theoretical extensions** (discussion papers)
2. Summarizes core findings with statistical validation
3. Presents the concrete "ask" (endorsement, feedback, collaboration)

What is Empirically Validated? (UTAC v2.0 Main Paper)

Core Findings

Dataset: 78 threshold systems across 5 scientific domains

Statistical Validation:

- **Domain clustering:** ANOVA $F(4,73)=185.3$, $p<10^{-20}$, $\eta^2=0.91$
- **Effect size:** Domain membership explains 91% of β-variance
- **Informational vs. Others:** $t(76)=14.2$, $p<10^{-20}$, Cohen's $d=2.1$ (very large effect)

Domain-Specific β-Bands

Domain	n	$\acute{\beta}$	Examples
Informational/ SOC	27	4.5 ± 0.9	LLMs, neuronal avalanches, markets
Geophysical	10	4.6 ± 0.8	Earthquakes, volcanic eruptions
Biological	18	7.4 ± 0.9	Microbiome, ecosystems
Climate	10	11.0 ± 1.0	AMOC, ice sheets,

Domain	n	$\acute{\beta}$	Examples
			permafrost
Neurodegenerative	20	13.0 ± 1.8	Huntington's, ALS, Alzheimer's

Interpretation

- **Informational Fixed Point:** $\beta \approx 4.2$ for high-dimensional, long-range coupled, mean-field-like processes (CCUC: Computational Criticality Universality Class)
- **Higher β in physical/biological systems:** Stronger material constraints, lower damping, steeper transitions
- **Golden Ratio scaling:** $\beta_n \approx \Phi^{n/3}$ predicts domain means with ¿8% error

Microfoundation

β emerges from coupling-to-noise ratio:

$$\beta \sim \alpha \frac{J}{T}$$

where:

- J = coupling strength (interaction energy)
- T = noise/damping (thermal fluctuations, stochasticity)
- $\alpha \approx 2$ = universal geometric factor (from Renormalization Group theory)

Physical interpretation:

- **Low β (≈ 4.5):** High damping, fast feedback, gradual transitions
- **High β (≈ 11):** Low damping, slow approach, catastrophic transitions
- **Extreme β (≈ 13):** Pathological hyper-steepness, essentially no warning

What is Theoretical/Hypothetical? (Discussion Papers)

Two Companion Papers

1. Dringende Klimakaskade – Seismische Gefahr

Type: Discussion paper (German)

Content: Climate-geophysical coupling risks

Status: Language toned down in corrected version; clearly marked as discussion paper

Key hypotheses:

- Climate-driven threshold crossings (WAIS collapse, AMOC shutdown) may trigger geophysical responses
- Mechanisms: isostatic rebound, methane hydrate destabilization, volcanic pressure release
- Warning: NOT empirically validated; requires specific datasets for falsification

2. TAC Type-6 – Implosive Genesis & Cubic-Root Jump

Type: Theoretical derivation

Content: Formal framework for extreme β-dynamics near $R \approx \Theta$

Status: Mathematically consistent, NOT empirically validated; clearly marked as theory

Key hypotheses:

- $\beta_{\text{dynamic}} \propto (R - \Theta)^{-1/3}$ near threshold (cubic-root singularity)
- Geometric interpretation: 3D parameter space → 1D threshold projection
- Testable prediction: Steepness spikes in high-resolution temporal data near known thresholds

Consistency Adjustments in This Corrected Version

1. β-Dynamics Near Threshold ($R \approx \Theta$)

Unified form: $\beta_{\text{dynamic}} \propto (R - \Theta)^{-1/3}$ (inverse cube root)

Intuition: 3D coupling space (R, Θ, β) undergoes volumetric compression → 1D threshold projection → divergence of steepness as $R \to \Theta$

Alternative forms: Square-root scaling ($\propto (R - \Theta)^{-1/2}$) discussed in appendices as sensitivity analysis

2. Tone in Policy-Adjacent Passages

Changes:

- Superlatives reduced (e.g., "unimaginable magnitude" → "potentially extreme magnitude")
- Confidence intervals and source references added where available
- Every policy statement explicitly labeled as recommendation/hypothesis

3. Separation of Empirical vs. Theoretical Claims

Clear demarcation:

- **Main Paper:** Only empirically validated claims
- **Discussion Papers:** Clearly marked as theoretical/hypothetical
- **This README:** Explicit separation in Sections 1 and 2

Concrete "Ask"

1. Feedback on Robustness of β-Clustering

Questions for reviewers:

- Are logistic fits (sigmoid) appropriate, or should we test alternative models (power-law with cutoffs, MLE)?
- Are bootstrap confidence intervals (1000 iterations, 95% CIs) sufficient for uncertainty quantification?
- Are there potential confounders we haven't considered (temporal resolution, measurement heterogeneity, domain classification)?

2. Comments on J/T Microfoundation

Questions for reviewers:

- Is the coupling-to-noise interpretation physically plausible across diverse domains?
- Are there domain-specific factors (e.g., material properties, biological constraints) that could explain β-variation better than J/T?
- Can this framework be reconciled with existing theories (RG, SOC, percolation)?

3. Critical Review of $(R-\Theta)^{-1/3}$ Form

Questions for reviewers:

- Is the cubic-root form mathematically justified, or are alternative scalings (square-root, logarithmic) more plausible?
- What empirical data would be needed to test this hypothesis rigorously?
- Are there existing datasets (high-resolution temporal measurements near thresholds) that could validate or falsify this claim?

4. Guidance on Suitable Datasets

Specific requests:

- **AMOC/WAIS:** Paleo-climate proxies with high temporal resolution
- **MEG/EEG Avalanches:** Neuronal activity time-series near consciousness thresholds (e.g., anesthesia transitions)
- **USGS Reanalyses:** Earthquake catalog data with strain gauge measurements near known faults
- **LLM Scaling Laws:** Parameter count vs. emergent capability data (GPT-5, Claude 4, Gemini Ultra)

5. arXiv Endorsement / Journal Submission Advice

Primary request: arXiv endorsement for `physics.data-an` or `cond-mat.stat-mech`

Alternative: If endorsement is not possible, guidance on direct journal submission strategy (targets: *Chaos*, *Physical Review E*, *Nature Communications*)

Citation & Contact

How to Cite

Main Paper:

> Römer, J. (2025). Domain-Specific Universality in Threshold Activation Criticality: A Multi-Attractor Framework. *Zenodo*. DOI: 10.5281/zenodo.17472834

BibTeX:

```
@misc{romer2025utac,
  author = {Römer, Johann},
  title = {Domain-Specific Universality in Threshold
    Activation Criticality},
  year = {2025},
  publisher = {Zenodo},
  doi = {10.5281/zenodo.17472834}
}
```

Contact Information

- **Primary:** Via Zenodo repository (DOI: 10.5281/zenodo.17472834)
- **Institutional collaborators:** Listed in cover sheet of each document

Document Structure Overview

Zenodo Repository Contents

1. **UTAC_v2_0_Main_Paper.pdf** (12 pages)
   - Empirical results only
   - Peer-review ready
   - Statistical validation, domain clustering, Golden Ratio scaling
2. **UTAC_Main_Paper_Cover_Sheet.pdf** (1 page)
   - Quick summary for reviewers
   - Highlights key sections and robustness checks
3. **UTAC_Executive_Summary.pdf** (2 pages)
   - Policy-focused brief
   - 5 key findings, 3 policy recommendations
4. **TAC_Type6_Seismic_Cascade.pdf** (11 pages)
   - Theoretical extension
   - Clearly marked as hypothetical
   - Validation roadmap included
5. **Klimakaskade_Seismische_Gefahr.pdf** (10 pages, German)
   - Discussion paper
   - Climate-geophysical coupling hypotheses
   - Toned down in corrected version
6. **UTAC_Dataset_Documentation.pdf** (3 pages)
   - CSV format specifications
   - Replication instructions
   - Quality control metrics
7. **Data folder:** 8 CSV files (78 systems total)
8. **Code folder:** R scripts for replication (optional)

Next Steps After Review

If Feedback is Positive

1. Address reviewer comments, revise as needed
2. Submit to arXiv (if endorsement received)
3. Submit to *Chaos* or *Physical Review E* for peer review
4. Expand validation datasets (LLM scaling, consciousness, paleo-seismic)

If Major Revisions Required

1. Implement robustness checks (alternative models, sensitivity analyses)
2. Expand sample size (target: 100–200 systems)
3. Strengthen microfoundation (more detailed RG derivation)
4. Consider domain-specific publications (Climate, AI Safety, Consciousness)

Final Note

Thank you for your time and consideration.
We look forward to your feedback.
Johann Römer | Independent Researcher, Marburg, Germany
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